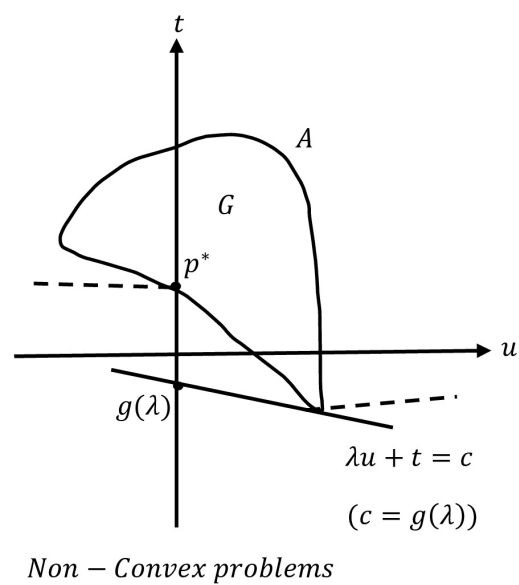
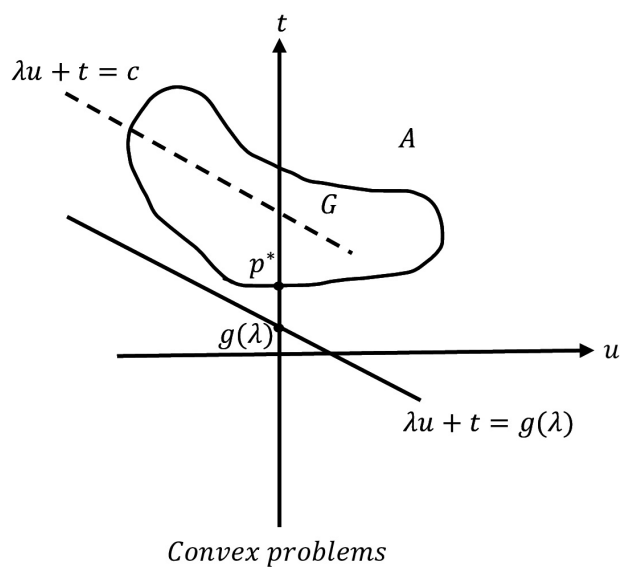


ECE 647: Lecture 30

1 Graphical Interpretation

- Consider the case with only inequality constraints
- Boyd P234
- Let $G = \{\underbrace{f_1(x), \dots, f_M(x)}_{u'}, \underbrace{f_0(x)}_{t'} \mid x \in D\}$
- Let $A = \{(u, t) \mid \exists x \in D, f_i(x) \leq u_i, f_0(x) \leq t\}$
 - like an epigraph, but without the variable x .
- If the problem is convex, then A is a convex set. (G may not be convex)
- Further, the lower boundary must be non-increasing in u .



- Let p^* denote the optimal value of the primal problem. Then

$$\begin{aligned} p^* &= \inf\{t' | (u', t') \in G, u' \leq 0\} \\ &= \inf\{t | (u, t) \in A, u = 0\} \end{aligned}$$

2 The Dual

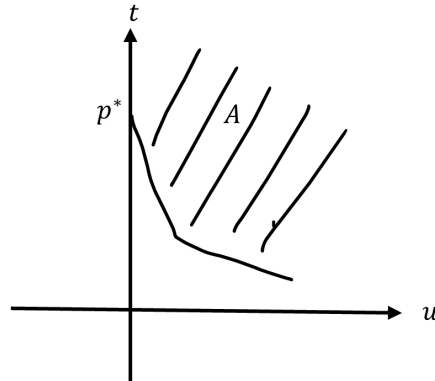
- Let

$$\begin{aligned} g(\lambda) &= \inf\{\lambda u' + t' | (u', t') \in G\} \\ &= \inf\{\lambda u + t | (u, t) \in A\}, \lambda \geq 0 \end{aligned}$$

- It is the intersection with the vertical-axis at the highest line below A .
- Clearly $p^* \geq g(\lambda)$
- If there is a hyperplane that supports A at the point $(., p^*)$, then

$$p^* = g(\lambda^*)$$

- The Slater condition ensures that there is some part of A on the left of the vertical axis $\Rightarrow$ supporting hyperplane must exist.
- If the Slater condition does not hold:



- To argue about this, we need an important result for convex sets, called separation theorem.
- It is quite technical, so we will focus on the implication of strong duality first, and return to separation theorem later.

3 Complimentary slackness

- We have seen some important properties of the pair of primal solutions x^* and dual solutions (λ^*, ν^*)

1. $f_i(x^*) \leq 0, h(x^*) = 0$
2. $\lambda^* \geq 0$
3. x^* minimizes $L(x, \lambda^*, \nu^*)$ over x

- Below the 4th important property.

- **Statement:** Let λ^*, ν^* be the optimal solution to the dual and x^* be the optimal solution to the primal. If the duality gap is zero, the

$$\lambda^* f_i(x^*) = 0 \quad \forall i$$

Further, x^* minimizes $L(x, \lambda^*, \nu^*)$.

- If $\lambda_i^* > 0$, then $f_i(x^*) = 0$
- If $f_i(x^*) < 0$, then $\lambda_i^* = 0$
- If the langrange multiplier is non-zero, then the constraint must be binding/tight.

Proof:

Assume $g(\lambda^*, \nu^*) = f_0(x^*)$. Since,

$$g(\lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*)$$

$$= f_0(x^*) + \sum \lambda_i^* f_i(x^*) + \sum \nu_i^* h_i(x^*)$$

$$\Rightarrow 0 \leq \sum \underbrace{\lambda_i^*}_{\geq 0} \underbrace{f_i(x^*)}_{\leq 0}$$

$$\Rightarrow \lambda_i^* f_i(x^*) = 0 \quad \forall i$$

Further, $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*) = f_0(x^*)(\#)$

Important Remarks:

- No convexity is assumed
- This holds when we pair any primal solution x^* with any dual solution λ^* !

These properties are useful when we want to find the primal solution after we found the dual solution.

- If the primal variable that minimizes $L(x, \lambda^*, \nu^*)$ is unique $\rightarrow$ Great!
- If not, need to find x^* that satisfies the primal constraints and the cs-consition.
 - If $\lambda_i^* > 0$, the corresponding constraint must be binding/tight.

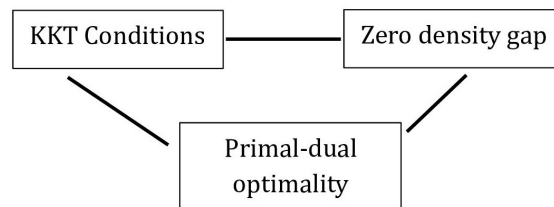
4 KKT (Karush-Kuhn-Tucker) conditions

A pair of primal-dual points (x^*, λ^*, ν^*) are said to satisfy the KKT condition if

- $f_i(x^*) \leq 0, h_i(x^*) = 0$ (satisfy the constraints)
- $\lambda_i^* \geq 0$ (valid lagrange multiplier)
- $\lambda_i^* f_i(x^*) = 0 \forall i$ (complementary slackness)
- x^* minimizes $f_0(x) + \sum_i \lambda_i^* f_i(x) + \sum_i \nu_i^* h_i(x)$ over all x . (min Lagrangian)

Note: If the functions are differentiable and convex, then this last condition is equivalent to

$$\nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) + \sum_i \nu_i^* \nabla h_i(x^*) = 0 \quad (*)$$



1. If the duality gap is zero, then any pair of primal and dual optimal points must satisfy KKT condition.

KKT $\leftarrow$ Zero duality-gap and primal & dual optimal

- No convexity is assumed.
 - follows from the complementary slackness condition.
2. If a pair of primal and dual points satisfy the KKT conditions, then they must be optimal and there is zero duality gap.

KKT $\rightarrow$ Zero duality-gap and primal - dual optimal

- We always have, for any feasible x & $\lambda \geq 0$:

$$g(\lambda, \nu) \leq L(x, \lambda, \nu) \leq f_0(x)$$

- If the KKT condition holds, then

$$g(\lambda^*, \nu^*) = L(x, \lambda^*, \nu^*) = f_0(x)$$

- Hence, the duality gap is zero and x^* and (λ^*, ν^*) are the primal and dual optimal points, respectively.
 - Convexity is needed only for the first order condition (*). Otherwise (*) may not imply that x^* minimizes the lagrangian.
3. If the Slater condition holds, and the primal problem is convex, then the dual optimal solution always exist, and the duality gap is zero.
 - Hence, the KKT condition provides a necessary and sufficient condition for optimality with constraints.

KKT $\xleftrightarrow{\text{Convex+Slater}}$ primal - dual optimal

- Further, the 4th condition in KKT is simplified to a gradient condition.
 - For non-convex problems, however, this simplified form is necessary but not sufficient.